

Abdullah Al Shadi

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Research interests

- ❖ Semiconductor Devices
- ❖ Novel Materials
- ❖ Nanotechnology
- ❖ Energy Harvesting
- ❖ Renewable Energy
- ❖ Device Fabrication
- ❖ VLSI Technology
- ❖ Embedded System
- ❖ Hardware Prototyping
- ❖ IoT Electronics

Education

Khulna University of Engineering & Technology, Bangladesh

BSc. in Electrical and Electronic Engineering

CGPA: 3.21 (out of 4)

Khulna, Bangladesh

Jan. 2019– Mar. 2024

Publications

- Journals.
- ❖ Muneef Hasan, et al. “A comprehensive analysis of structural, electronic, optical, mechanical, thermodynamic and thermoelectric properties of direct band gap Sr3BF3 (B = As, Sb) photovoltaic compounds: DFT-GGA and mBJ approach”, *Inorganic Chemistry Communications, Volume 171*, January 2025, 113607. (**IF :4.4**) ([View](#))
 - ❖ “Comprehensive Investigation of Strain-Induced Transformations in Lead- Free FrSnA3 (A = Cl, Br, and I) Perovskite: Unraveling Structural, Electronic, Optical, and Mechanical Properties through DFT Calculations”, 2025, [*Manuscript under review*].
- Conferences.
- ❖ Abdullah Al Shadi, et al. “Investigation of the impacts of different parameters of Buffer Carbon Trapping in AlGaIn/GaN HEMTs”, *4th International Conference on Electrical, Computer and Communication Engineering (ECCE)*, 13-15 February 2025. ([View](#))
 - ❖ Bijoy Sorker, et al. “First-Principles Investigation of the Structural, Electronic, Mechanical, and Optical Properties of A3NBr3 (A = Ca, Sr, and Ba)”, *4th International Conference on Electrical, Computer and Communication Engineering (ECCE)*, 13-15 February 2025. ([View](#))

Projects

- ❖ **USB-C to UART Converter (KiCad)**
 - ◆ Designed a compact USB-C to UART converter with ESD protection and voltage level shifting using FT-332 IC. Completed schematic, footprint creation, PCB layout, and 3D visualization using KiCad.
- ❖ **DC Motor Speed Controller (KiCad)**
 - ◆ Developed a PWM-based speed controller using a 555 timer and transistor driver. Executed schematic design, custom footprints, and PCB layout with real-time speed control via a potentiometer.
- ❖ **Servo Motor Tester (KiCad)**
 - ◆ Created a standalone PWM generator for servo testing using a 555 timer IC. Handled full schematic design, PCB layout, and 3D model generation in KiCad.
- ❖ **AC to DC Power Supply (KiCad)**
 - ◆ Designed a regulated power supply converting 230V AC to 5V/12V DC. Completed end-to-end schematic, footprint design, PCB layout, and safety-focused component selection in KiCad.
- ❖ **DC Motor Control using Full Bridge Power Converter for Locomotive Train Machine**
 - ◆ Implemented a DC motor control system using a power electronics-based H-bridge converter to regulate speed and direction. Integrated microcontroller-based control, highlighting real-world application of electric vehicle drive systems.
- ❖ **IOT based voice controlled smart home automation**
 - ◆ Developed a voice-activated home automation system using a voice assistant to control household appliances, enabling hands-free interaction and enhancing user convenience.
- ❖ **Designing a Single Phase, Shell Type Transformer**
 - ◆ Collaboratively designed and built a 220V/12V, 400VA single-phase transformer. Achieved practical implementation of transformer theory with results closely matching design specifications.

- ❖ **Flat Wiring Design with Cost Estimation using AutoCAD Software**
 - ◆ *Designed a floor plan and performed conduit wiring layout using AutoCAD. Calculated the required illumination and appliance placement with a cost-effective wiring estimate of BDT 82,298.*
- ❖ **Autonomous Line Follower Robot**
 - ◆ *It's an Arduino-based project. The robot can follow a given path by avoiding all barriers and reaching the final destination. It is mainly a competition-based project to reach the final destination as early as possible.*

Software Skills

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|--|---|
| ❖ Programming Languages: Python, C, C++, MATLAB | ❖ Software and Tools: AutoCAD, Simulink, Proteus, TCAD, PSpice, KiCad, SCAPS |
| ❖ Hardware: PLC, Arduino IDE, STM32 | ❖ Web Technologies: HTML, CSS |
| ❖ Other Skills: Circuit Design, PCB Layout, Signal Processing, Embedded Systems | ❖ Office Tools: Word, Excel, PowerPoint |
| ❖ Designing Tools: Adobe Illustrator | |

Experiences

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| ❖ Executive-Projects & Product Development, Super Star Group | May.2025-Present |
| <ul style="list-style-type: none"> ◆ <i>Design and implement PCB layout of rechargeable Fan</i> ◆ <i>Develop the existing product quality</i> ◆ <i>Find the way of cost minimization</i> | |
| ❖ Organizing Secretary, EEE Makers Hub | Jan. 2023– Mar. 2024 |
| <ul style="list-style-type: none"> ◆ <i>Organized technical events, workshops, and seminars to promote skill development.</i> ◆ <i>Led technical initiatives, demonstrating strong organizational and leadership skills.</i> ◆ <i>Established connections with industry professionals for potential collaborations and opportunities.</i> | |

Awards and Achievements

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|---|------|
| ❖ Education Board Scholarship for a good result in Higher Secondary Education | 2018 |
| ❖ Regional Champion of National Creative Talent Hunt (Science) | 2015 |
| ❖ Education Board Scholarship for a good result in JSC Examination | 2013 |

Certifications

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|---|----------------|
| ❖ 30 Days Masterclass on PLC | November, 2024 |
| ❖ MATLAB Programming for Engineers and Scientists | October, 2022 |
| ❖ Mastering Programming with MATLAB | June, 2022 |
| ❖ Introduction to Programming with MATLAB | May, 2022 |
| ❖ Python For Beginners Course In-Depth | June, 2022 |

Language skills

- ❖ English (IELTS Band -7)
- ❖ Bangla (Native)

Reference

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